Lab Setup Instructions

Applied Agentic AI - Foundation to Production (US Track)

Walmart Global Tech Academy

For Lab Vendor Use Only

Purpose

This document provides step-by-step instructions for setting up the lab environment for a 5-day corporate AI training program. Follow each step in order and do not skip any step. Once setup is complete on one machine, replicate it on all participant machines.

Program details: 100 participants | 5 days | 5 sessions x 4 hours

Section 1: Machine Requirements

Each participant machine must meet the following minimum specifications.

Requirement	Minimum Specification
Processor	Intel Core i5 8th gen or newer | AMD Ryzen 5 3000 series or newer | Apple M1, M2, or M3
RAM	16 GB minimum (32 GB recommended for multi-agent workflows)
Free Storage	At least 10 GB of free disk space
Internet	Stable broadband connection (required for OpenAI API calls during all 5 sessions)
Display	1920 x 1080 resolution or higher

Supported Operating Systems:

Operating System	Supported Version
macOS	Ventura (13.x) | Sonoma (14.x) | Sequoia (15.x)
Windows	Windows 10 64-bit (version 21H2 or later) | Windows 11 64-bit (any edition)

Note: Linux is out of scope for this lab. If any machine is running Linux, contact the course instructor before proceeding.

Section 2: Before You Begin

Confirm the following before starting the setup steps:

- Python 3.10, 3.11, or 3.12 is available on each machine. Instructions for installing Python are provided in Step 4.
- Each machine has a working internet connection that can reach pypi.org to download Python packages.
- The API keys have been received from the course instructor. If keys have not yet been provided, create the .env file with blank values as described in Step 2. The instructor will fill in the values on Day 1 of the training.

Note: API keys are confidential. They must be stored only in the .env file inside the course folder and must never be copied into notebooks or shared outside the lab environment.

Step 1: Create the Course Folder on the Desktop

Create a folder named "Applied Agentic AI" on the Desktop of each participant machine. The folder name must be typed exactly as shown, with the same spaces and capital letters.

macOS

1. Click on the Desktop to bring it into focus.
2. Right-click (or two-finger click on a trackpad) on an empty area of the Desktop and select New Folder.
3. Type the folder name exactly as shown below and press Return:

```
Applied Agentic AI
```

Windows

1. Right-click on an empty area of the Desktop.
2. Select New > Folder from the menu.
3. Type the folder name exactly as shown below and press Enter:

```
Applied Agentic AI
```

Note: The folder name must be "Applied Agentic AI" with spaces. Do not use underscores or hyphens. The notebooks load files from this exact folder name.

Step 2: Create the .env File

Inside the "Applied Agentic AI" folder, create a plain text file named ".env" (with a dot at the beginning and no extension after it). This file is where API keys are stored. Leave all values blank for now.

macOS

Open Terminal by pressing Command + Space, typing Terminal, and pressing Enter. Then run the commands below. Copy and paste the entire block at once:

```
cd ~/Desktop/"Applied Agentic AI"
cat > .env << 'EOF'
OPENAI_API_KEY=
HF_TOKEN=
SERPAPI_API_KEY=
OPENWEATHERMAP_API_KEY=
NEWS_API_KEY=
TAVILY_API_KEY=
```

```
EOF
```

Verify the file was created:

```
ls -la
```

You should see ".env" listed in the output. The dot at the start of the filename means it is a hidden file, which is correct.

Windows

Open Command Prompt by pressing Windows Key + R, typing cmd, and pressing Enter. Then run the two commands below, one at a time:

```
cd "%USERPROFILE%\Desktop\Applied Agentic AI"
(echo OPENAI_API_KEY=& echo HF_TOKEN=& echo SERPAPI_API_KEY=& echo
OPENWEATHERMAP_API_KEY=& echo NEWS_API_KEY=& echo TAVILY_API_KEY=) > .env
```

Verify the file was created:

```
dir /a
```

You should see ".env" listed. The /a flag is required because .env is a hidden file on Windows.

For reference, the final .env file must contain exactly these six lines. Values will be filled in by the instructor on Day 1:

```
OPENAI_API_KEY=
HF_TOKEN=
SERPAPI_API_KEY=
OPENWEATHERMAP_API_KEY=
NEWS_API_KEY=
TAVILY_API_KEY=
```

Step 3: Create the requirements.txt File

Inside the "Applied Agentic AI" folder, create a file named "requirements.txt". This file lists all the Python packages needed to run every notebook across all 5 days of training, including the Day 5 capstone project.

macOS

In Terminal, first make sure you are inside the course folder:

```
cd ~/Desktop/"Applied Agentic AI"
```

Then run the following command to create requirements.txt. Copy and paste the entire block at once:

```
cat > requirements.txt << 'EOF'
# ============================================================
# Applied Agentic AI - Foundation to Production
# Walmart Global Tech Academy  |  requirements.txt
# ============================================================

# Core OpenAI SDK
openai>=1.30.0
tiktoken>=0.7.0

# Environment variables
python-dotenv>=1.0.0

# LangChain ecosystem (Days 1-5)
langchain>=0.2.16
langchain-core>=0.2.38
langchain-openai>=0.1.22
langchain-community>=0.2.16

# LangGraph - stateful agent framework (Days 2-5)
langgraph>=0.2.0

# Vector database - RAG notebooks (Days 1 and 5)
chromadb>=0.5.0

# FastAPI stack - Day 4 Notebook 11
fastapi>=0.111.0
uvicorn[standard]>=0.30.0
httpx>=0.27.0

# Utilities
pydantic>=2.7.0
rich>=13.7.0
requests>=2.31.0

# Jupyter environment
jupyter>=1.0.0
ipykernel>=6.29.0
```

```
notebook>=7.2.0
EOF
```

Windows

Open Notepad by pressing the Windows Key, typing Notepad, and pressing Enter.

Copy the text below and paste it into Notepad:

```
# ============================================================
# Applied Agentic AI - Foundation to Production
# Walmart Global Tech Academy  |  requirements.txt
# ============================================================

# Core OpenAI SDK
openai>=1.30.0
tiktoken>=0.7.0

# Environment variables
python-dotenv>=1.0.0

# LangChain ecosystem (Days 1-5)
langchain>=0.2.16
langchain-core>=0.2.38
langchain-openai>=0.1.22
langchain-community>=0.2.16

# LangGraph - stateful agent framework (Days 2-5)
langgraph>=0.2.0

# Vector database - RAG notebooks (Days 1 and 5)
chromadb>=0.5.0

# FastAPI stack - Day 4 Notebook 11
fastapi>=0.111.0
uvicorn[standard]>=0.30.0
httpx>=0.27.0

# Utilities
pydantic>=2.7.0
rich>=13.7.0
requests>=2.31.0

# Jupyter environment
jupyter>=1.0.0
ipykernel>=6.29.0
notebook>=7.2.0
```

In Notepad, go to File > Save As. Navigate to Desktop > Applied Agentic AI. Set the File name to requirements.txt and set Save as type to All Files. Click Save.

Note: If Save as type is left as Text Documents, the file will be saved as requirements.txt.txt and pip will not find it. Always select All Files in the Save As dialog on Windows.

Step 4: Install Python 3.11

Python 3.10, 3.11, or 3.12 is required. Python 3.13 is not yet supported by some packages used in this training. If installing Python fresh on a machine, install Python 3.11.

macOS - Check existing installation

Open Terminal and run:

```
python3 --version
```

If the output shows Python 3.10.x, 3.11.x, or 3.12.x, Python is already installed. Skip to Step 5.

If Python is missing or the version is outside the supported range, download the macOS installer from:

```
https://www.python.org/downloads/macos/
```

Download the latest Python 3.11 release. Open the downloaded .pkg file and follow the on-screen instructions to install.

Windows - Check existing installation

Open Command Prompt and run:

```
python --version
```

If the output shows Python 3.10.x, 3.11.x, or 3.12.x, Python is already installed. Skip to Step 5.

If Python is missing or the version is wrong, download the 64-bit Windows installer from:

```
https://www.python.org/downloads/windows/
```

During installation, follow these steps exactly:

1. At the first installer screen, check the box that says "Add Python to PATH". This step is required. Without it, Python commands will not work in Command Prompt.
2. Click Install Now and wait for the installation to complete.
3. Close the installer. Open a new Command Prompt window (not the existing one) and run python --version to confirm the installed version.

Step 5: Create the Virtual Environment (wal-env)

A virtual environment keeps all packages for this training isolated from other Python software on the machine. The environment must be named "wal-env". These steps must be completed inside the course folder.

macOS Instructions

1. Open Terminal.
2. Navigate to the course folder:

```
cd ~/Desktop/"Applied Agentic AI"
```

3. Create the virtual environment:

```
python3 -m venv wal-env
```

A folder named "wal-env" will be created inside the course folder. This takes 10 to 20 seconds.

4. Activate the virtual environment:

```
source wal-env/bin/activate
```

The terminal prompt will change to show "(wal-env)" at the beginning. This confirms the environment is active. All pip and python commands now operate inside wal-env.

5. Upgrade pip to the latest version:

```
pip install --upgrade pip
```

6. Install all required packages from requirements.txt:

```
pip install -r requirements.txt
```

This step downloads and installs approximately 30 packages and their dependencies. It will take between 5 and 15 minutes depending on internet speed. Wait for the process to finish completely before moving to the next step. You will see the prompt return when it is done.

7. Register wal-env as a Jupyter kernel:

```
python3 -m ipykernel install --user --name=wal-env --display-name="Python (wal-
env)"
```

This step makes "Python (wal-env)" appear in the Jupyter kernel selector so participants can run notebooks in the correct environment.

8. Confirm packages installed correctly:

```
pip show openai langchain langgraph chromadb fastapi rich
```

You should see name and version details printed for each of the six packages above. If any package shows an error, re-run the pip install step from Step 5.5.

Windows Instructions

1. Open Command Prompt by pressing Windows Key + R, typing cmd, and pressing Enter.
2. Navigate to the course folder:

```
cd "%USERPROFILE%\Desktop\Applied Agentic AI"
```

3. Create the virtual environment:

```
python -m venv wal-env
```

A folder named "wal-env" will be created inside the course folder. This takes 10 to 20 seconds.

4. Activate the virtual environment:

```
wal-env\Scripts\activate
```

The Command Prompt will show "(wal-env)" at the beginning. This confirms the environment is active.

Note: If you see an error saying "running scripts is disabled on this system", open PowerShell as Administrator, run the command below, close PowerShell, and then try the activation command again in Command Prompt:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

5. Upgrade pip:

```
python -m pip install --upgrade pip
```

6. Install all required packages:

```
pip install -r requirements.txt
```

This step downloads and installs approximately 30 packages. It will take between 5 and 15 minutes. Wait for the process to complete before proceeding.

7. Register wal-env as a Jupyter kernel:

```
python -m ipykernel install --user --name=wal-env --display-name="Python (wal-
env)"
```

8. Confirm packages installed correctly:

```
pip show openai langchain langgraph chromadb fastapi rich
```

You should see name and version details for each package. If any shows an error, re-run pip install from Step 5.5.

Step 6: Verify the Setup

Run the following verification test on every participant machine before the training begins. This takes less than 2 minutes per machine.

Activate the environment (if not already active)

macOS - open Terminal and run:

```
cd ~/Desktop/"Applied Agentic AI"
source wal-env/bin/activate
```

Windows - open Command Prompt and run:

```
cd "%USERPROFILE%\Desktop\Applied Agentic AI"
wal-env\Scripts\activate
```

Run the import test (macOS and Windows)

With "(wal-env)" visible in the prompt, start Python:

```
python
```

Then type or paste each line below and press Enter after each one:

```
import openai; print('openai OK')
import langchain; print('langchain OK')
import langgraph; print('langgraph OK')
import chromadb; print('chromadb OK')
import fastapi; print('fastapi OK')
import rich; print('rich OK')
import dotenv; print('python-dotenv OK')
print('All imports successful')
exit()
```

Expected output:

```
openai OK
langchain OK
langgraph OK
chromadb OK
fastapi OK
rich OK
python-dotenv OK
```

```
All imports successful
```

If all 8 lines print without any errors, the machine is ready for training.

Verify Jupyter Notebook

With the environment still active, run:

```
jupyter notebook
```

A browser window will open showing the Jupyter file browser. Click New (top-right corner) and verify that "Python (wal-env)" appears in the list of available kernels. Once confirmed, close the browser tab and press Ctrl + C in the terminal to stop the Jupyter server.

Note: If "Python (wal-env)" does not appear in the kernel list, re-run the ipykernel install command from Step 5.6 and then restart Jupyter.

Expected Folder Structure After Setup

After completing all 6 steps, the course folder on the Desktop should contain the following:

```
Desktop/
  Applied Agentic AI/
    .env                    (hidden file - stores API keys)
    requirements.txt        (list of packages)
    wal-env/                (virtual environment)
      bin/                  (macOS: Python interpreter and scripts)
      Scripts/              (Windows: Python interpreter and scripts)
      lib/                  (installed packages)
```

Note: Course notebooks will be distributed by the instructor at the start of Day 1. Do not copy any notebook files to participant machines in advance.

Quick Reference: Daily Activation

Participants will need to activate the wal-env environment at the start of each training day. The commands are provided below. Share this table with participants on Day 1.

OS	Command to Activate wal-env
macOS	`source ~/Desktop/"Applied Agentic AI"/wal-env/bin/activate`

Windows	`"%USERPROFILE%\Desktop\Applied Agentic AI\wal-env\Scripts\activate"`

To deactivate the environment at the end of a session (both macOS and Windows):

```
deactivate
```

Contact

If you encounter any problem during setup that is not addressed in this guide, contact the course instructor before making any changes:

Dr. Darshan Ingle | ingledarshanlabs@gmail.com

Do not modify the setup, install additional packages, or change any configuration without approval from the instructor.